

Project Design Phase Solution Architecture

Date	9 June 2026
Team ID	
Project Name	ShopEZ
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

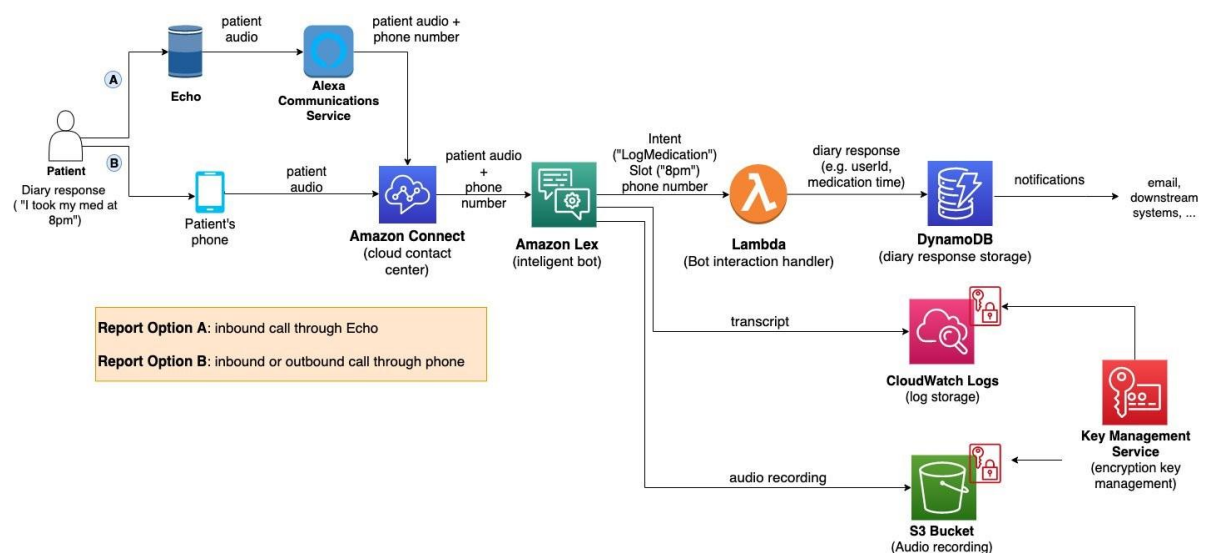


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

ShopEZ - Solution Architecture

Frontend (React.js):

The user interacts with the React frontend running on localhost:3000. It has pages for Login, Register, Home, Cart, Orders and Admin. Tailwind CSS is used for styling. Axios is used to make API calls to the backend.

Backend (Node.js + Express.js):

The Express server runs on localhost:5000. It handles all API routes for authentication, products, cart and orders. JWT is used for authentication and bcryptjs is used for password hashing.

Database (MongoDB):

MongoDB runs locally and stores four collections - users, products, carts and orders. Mongoose is used to define schemas and interact with the database.

How they connect:

1. User opens the React app in the browser
2. React sends HTTP requests to the Express API using Axios
3. Express processes the request and queries MongoDB using Mongoose
4. MongoDB returns the data to Express
5. Express sends the response back to React
6. React displays the data to the user

Collections:

- users: stores name, email, hashed password and role
- products: stores name, price, category, stock, image and rating
- carts: stores user reference and array of products with quantity
- orders: stores user reference, items, total amount and status